CSE 4309: Theory of Computing

Lecture 3 — Non-Regular Languages & the Pumping Lemma

Syed Rifat Raiyan
Lecturer
Department of Computer Science & Engineering
Islamic University of Technology, Dhaka, Bangladesh
Email: rifatraiyan@iut-dhaka.edu

Lecture Plan

The agenda for today

- What is a Non-Regular Language?
- What is the Pumping Lemma?
- How can we use the pumping lemma to prove non-regularity?
- Example proofs

Non-Regular Languages

What are they?

Non-regular languages are formal languages that ***cannot*** be described by a regular expression or recognized by a Finite Automaton (FA) because they require infinite memory to count or match patterns.

How do we show a language is not regular?

- Remember, to show a language *is* regular, we give a DFA.
- To show a language is *not* regular, we must give a proof.
- It is ***not enough*** to say that you couldn't find a DFA for it, therefore the language isn't regular.

Two examples: Here $\Sigma = \{0,1\}$.

1. Let $B = \{w \mid w$ has equal numbers of 0s and 1s$\}$

Intuition: B is not regular because DFAs cannot count unboundedly.

2. Let $C = \{w \mid w$ has equal numbers of 01 and 10 substrings$\}$

$01\underline{01} \notin C \qquad 01\underline{10} \in C$

Intuition: C is not regular because DFAs cannot count unboundedly.
However C is regular! Sometimes intuition works, but it can also be wrong.

Moral: You need to give a proof.

Pumping Lemma

Method for proving non-regularity

Pumping Lemma: For every regular language A, there is a number p (the "pumping length") such that if $s \in A$ and $|s| \geq p$ then $s = xyz$ where

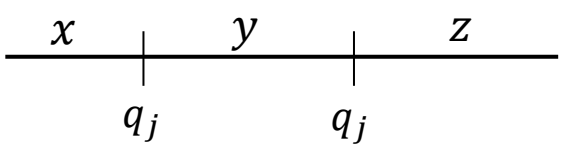
1) $xy^iz \in A$ for all $i \geq 0$
2) $y \neq \varepsilon$
3) $|xy| \leq p$

$$y^i = \underbrace{yy \cdots y}_{i}$$

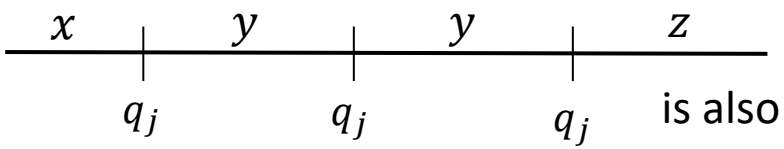
Informally: A is regular → every long string in A can be pumped and the result stays in A.

Proof: Let DFA M recognize A. Let p be the number of states in M. Pick $s \in A$ where $|s| \geq p$.

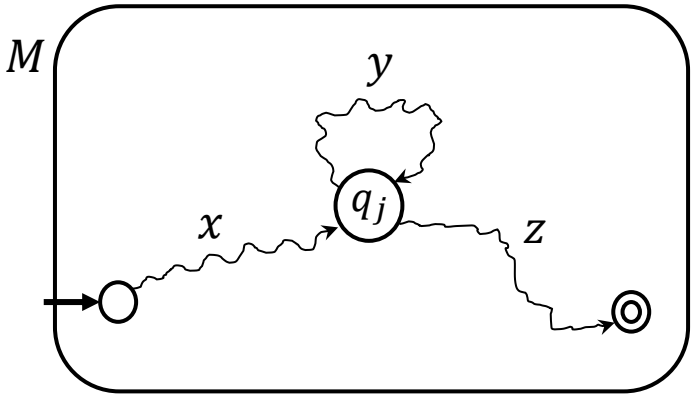
$s =$



M will repeat a state q_j when reading s because s is so long.



is also accepted



The path that M follows when reading s.

Proving Non-Regularity

An example

Pumping Lemma: For every regular language A, there is a p such that if $s \in A$ and $|s| \geq p$ then $s = xyz$ where

1) $xy^iz \in A$ for all $i \geq 0$ $\qquad y^i = yy \cdots y$
2) $y \neq \varepsilon$
3) $|xy| \leq p$

Let $D = \{0^k 1^k \mid k \geq 0\}$

Show: D is not regular

Proof by Contradiction:

Assume (to get a contradiction) that D <u>is</u> regular.

The pumping lemma gives p as above. Let $s = 0^p 1^p \in D$.

Pumping lemma says that can divide $s = xyz$ satisfying the 3 conditions.

$$s = \underbrace{000 \cdots 000111 \cdots 111}_{\overbrace{x \mid y}^{\leftarrow\ \leq p\ \rightarrow} \mid z}$$

But $xyyz$ has excess 0s and thus $xyyz \notin D$ contradicting the pumping lemma.

Therefore our assumption (D is regular) is false. We conclude that D is not regular.

Proving Non-Regularity

An example

Pumping Lemma: For every regular language A, there is a p such that if $s \in A$ and $|s| \geq p$ then $s = xyz$ where

1) $xy^iz \in A$ for all $i \geq 0$ $\qquad y^i = yy \cdots y$
2) $y \neq \varepsilon$
3) $|xy| \leq p$

Let $F = \{ww \mid w \in \Sigma^*\}$. Say $\Sigma^* = \{0,1\}$.

Show: F is not regular

$$s = \underbrace{000 \cdots 000000 \cdots 000}_{\overset{x}{\leftarrow}\ \overset{y}{\leq p}\ \overset{z}{\rightarrow}}$$

$y = 00$

Proof by Contradiction:

Assume (for contradiction) that F <u>is</u> regular.

The pumping lemma gives p as above. Need to choose $s \in F$. Which s?

Try $s = 0^p 0^p \in F$.

$$s = \underbrace{000 \cdots 001000 \cdots 001}_{\overset{x}{\leftarrow}\ \overset{y}{\leq p}\ \overset{z}{\rightarrow}}$$

Try $s = 0^p 1 0^p 1 \in F$. Show cannot be pumped $s = xyz$ satisfying the 3 conditions.

$xyyz \notin F$ Contradiction! Therefore F is not regular.

Proving Non-Regularity

An example

Variant: Combine closure properties with the Pumping Lemma.

Let $B = \{w \mid w$ has equal numbers of 0s and 1s$\}$

Show: B is not regular

Proof by Contradiction:

Assume (for contradiction) that B <u>is</u> regular.

We know that 0^*1^* is regular so $B \cap 0^*1^*$ is regular (closure under intersection).

But $D = B \cap 0^*1^*$ and we already showed D is not regular. Contradiction!

Therefore our assumption is false, so B is not regular.